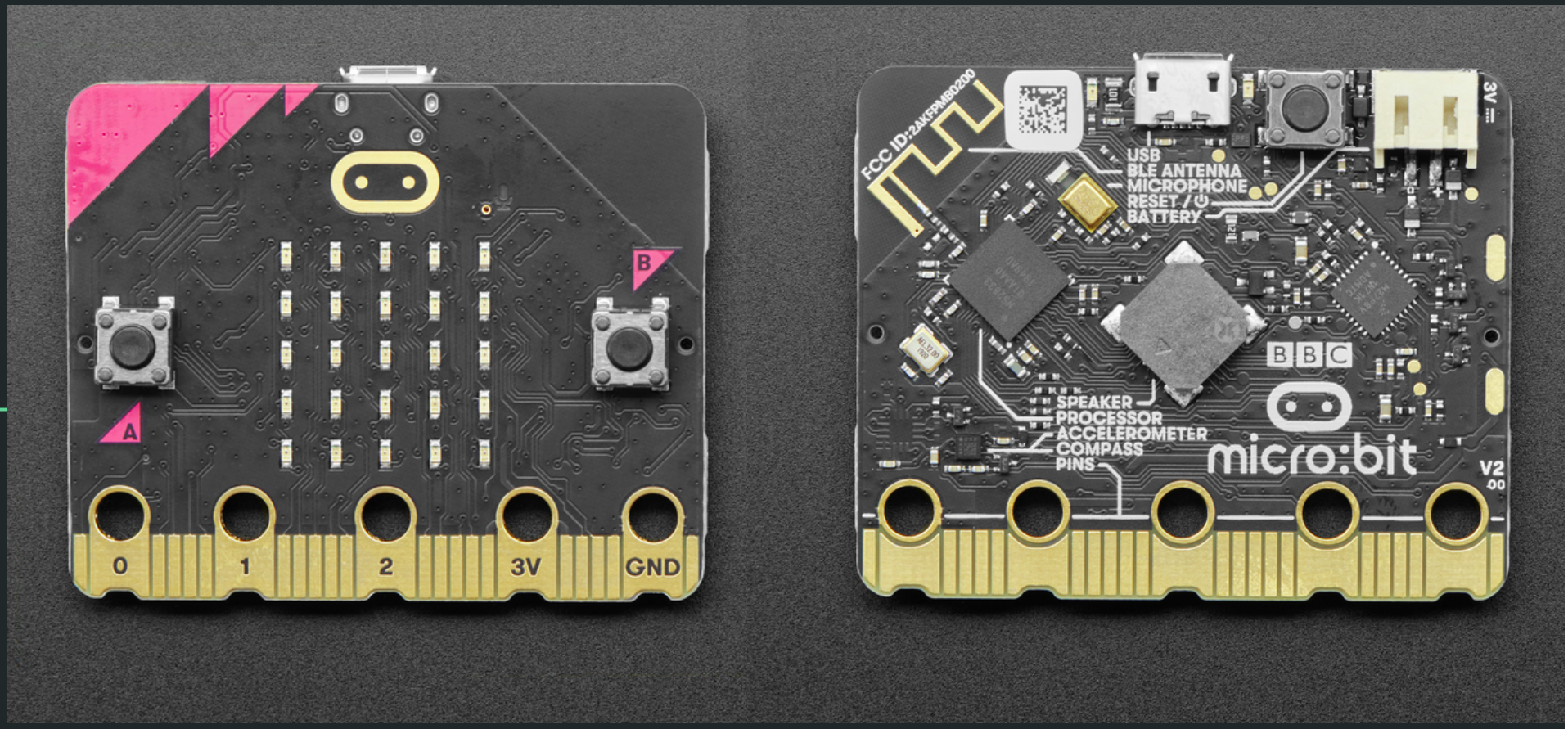
IEEE STEM Learning
BBC micro:bit v3

Steve Case

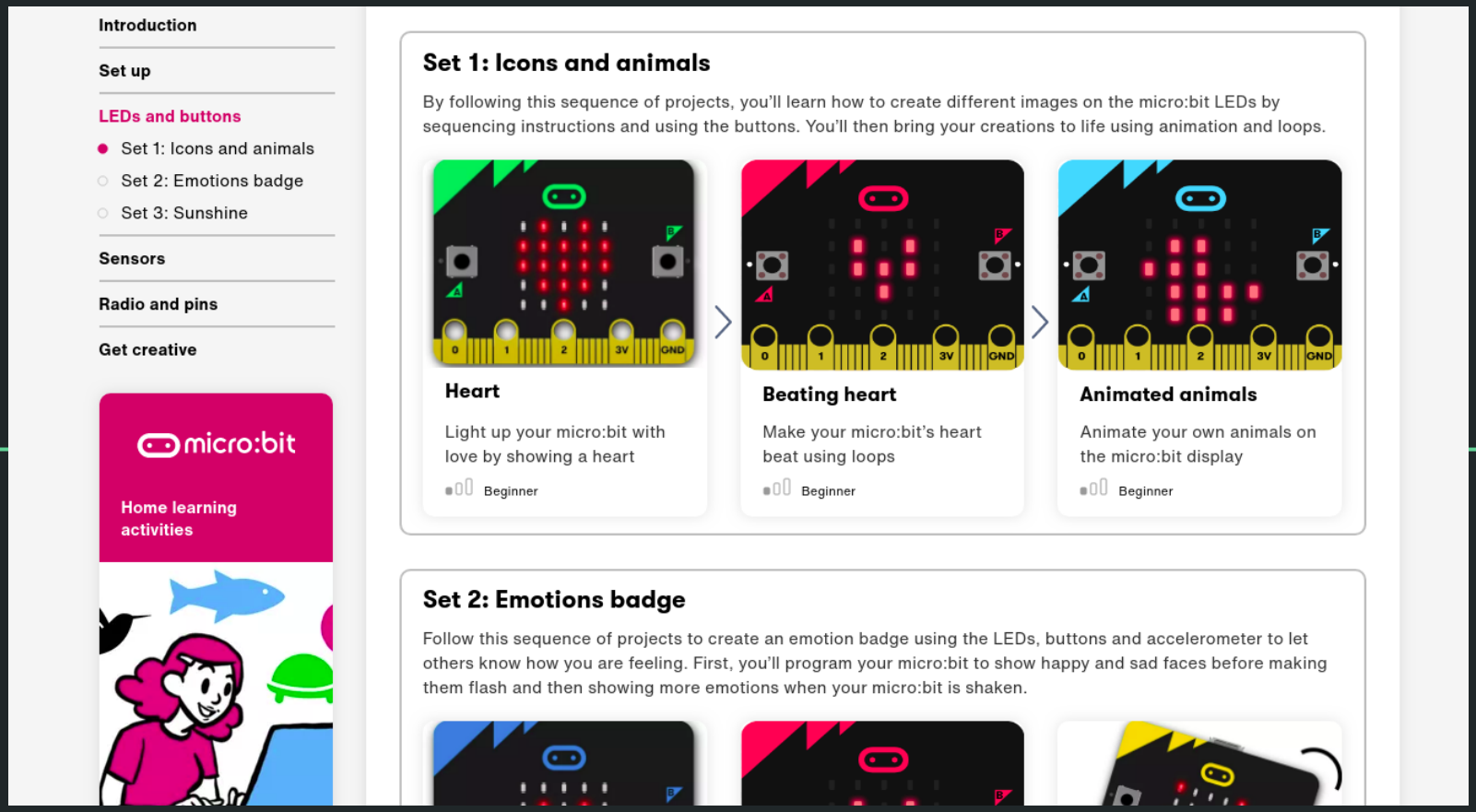
BBC micro:bit v2

BBC Micro:bit v2 Development Board



BBC micro:bit v1

BBC micro:bit



BBC Micro:bit

Set up your classroom

Begin your classroom set-up below. Once you've launched your classroom session you will have the option to add your own code to the classroom editor to share with your students.

Name your activity

Give your new activity a name...

Choose a programming language

MakeCode Python

Select storage setting

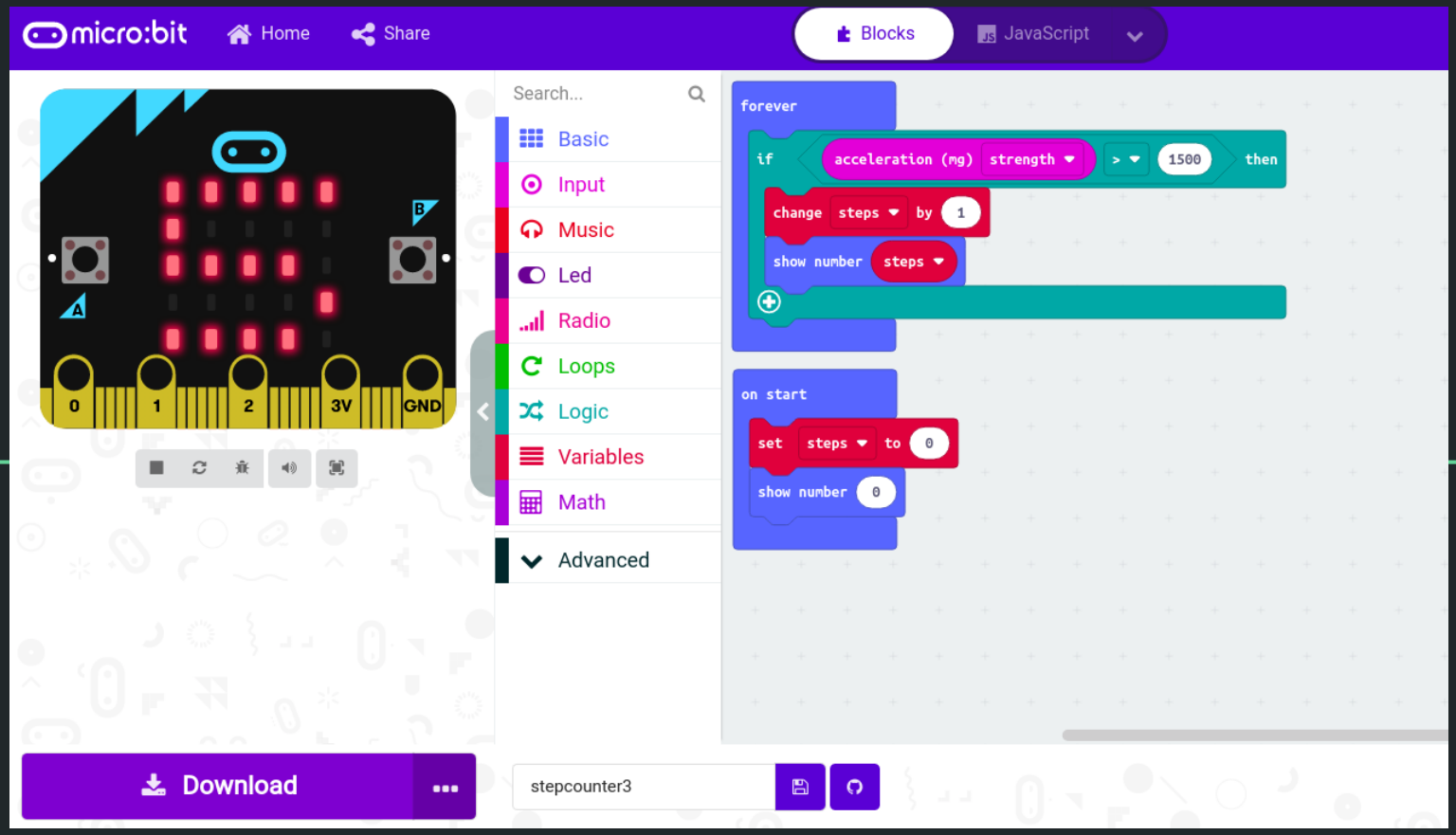
Use temporary local storage

Launch classroom

Explore other activities

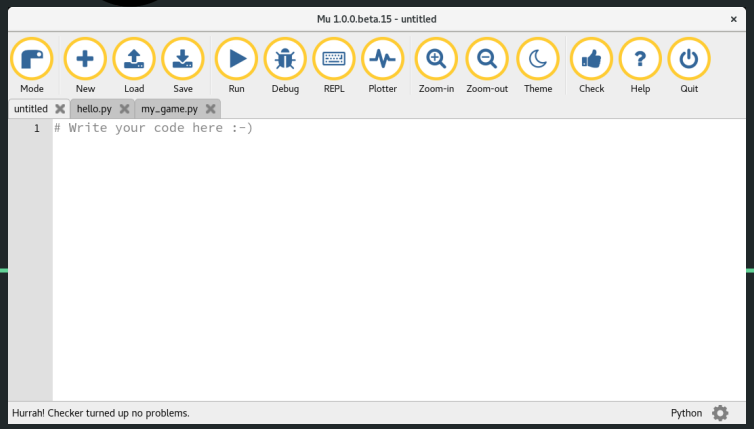
Want to resume an activity?

BBC micro:bit
Microsoft MakeCode

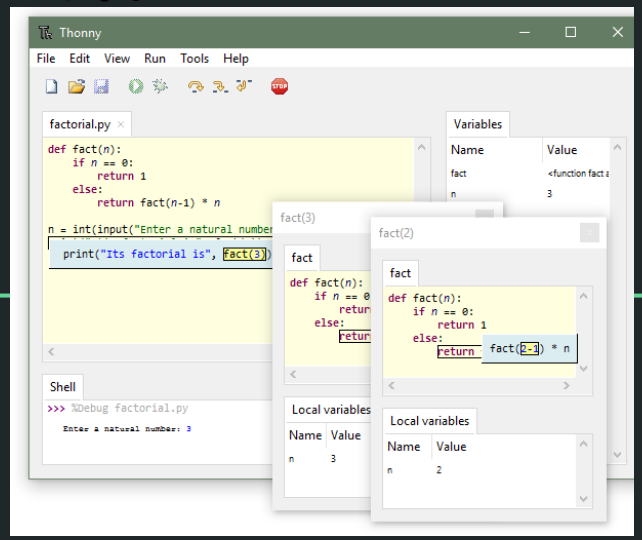


Python Editors

Mu Editor



Thonny



Python Editor Thonny

File Edit View Run Tools Help

Program tabs

Files

This computer
/home/steve/microbit/
students

- robot.py
- sound2.py
- speech.py
- temperature1.py
- temperature2.py
- weather_station.py
- while_loop1.py

micro:bit

- 8ball_1.py
- disp_n.py
- eight_ball.py
- for_loop1.py
- main.py

dice.py [main.py]

```
#
#  For loop demo with display
#

from microbit import *
import random
display.clear()


while True:
    for i in range(0, 5):
```

Shell Exception Program tree TODO

```
>>> %Run -c $EDITOR_CONTENT

 MPY: soft reboot
 Traceback (most recent call last):
   File "<stdin>", line 20, in <module>
 KeyboardInterrupt:
>>>
```

REPL

MicroPython (BBC micro:bit) • BBC micro:bit CMSIS-DAP @ /dev/ttyACM0

Images - import microbit, dir(Image)

'ALL_ARROWS', 'ALL_CLOCKS', 'ANGRY', 'ARROW_E', 'ARROW_N', 'ARROW_NE', 'ARROW_NW', 'ARROW_S', 'ARROW_SE', 'ARROW_SW', 'ARROW_W', 'ASLEEP', 'BUTTERFLY', 'CHESSBOARD', 'CLOCK1', 'CLOCK10', 'CLOCK11', 'CLOCK12', ' CLOCK2', 'CLOCK3', 'CLOCK4', CLOCK5', 'CLOCK6', CLOCK7', 'CLOCK8', CLOCK9', 'CONFUSED', 'COW', 'DIAMOND', 'DIAMOND_SMALL', 'DUCK', 'FABULOUS', 'GHOST', 'GIRAFFE', 'HAPPY', 'HEART', 'HEART_SMALL', 'HOUSE', 'MEH', 'MUSIC_CROTCHET', 'MUSIC_QUAVER', 'MUSIC_QUAVERS', 'NO', 'PACMAN', 'PITCHFORK', 'RABBIT', 'ROLLERSKATE', 'SAD', 'SCISSORS', 'SILLY', 'SKULL', 'SMILE', 'SNAKE', 'SQUARE','SQUARE_SMALL" STICKFIGURE', 'SURPRISED', 'SWORD', 'TARGET', 'TORTOISE', 'TRIANGLE', 'TRIANGLE_LEFT', 'TSHIRT', 'UMBRELLA', 'XMAS', 'YES',

Music - import music, dir(music)

'BADDY', 'BA_DING',
'BIRTHDAY', 'BLUES',
'CHASE', 'DADADADUM',
'ENTERTAINER',
'FUNERAL', 'FUNK',
'JUMP_DOWN', 'JUMP_UP',
'NYAN', 'ODE',
'POWER_DOWN',
'POWER_UP', 'PRELUDE',
'PUNCHLINE', 'PYTHON',
'RINGTONE',
'WAWAWAWAA',
'WEDDING', 'get_tempo

Reference Material

BBC micro:bit www.microbit.org

IEEE Course Material & Programs
https://github.com/IEEE-BV/microbit

Thonny-Editor Installation:
https://thonny.org/

Adafruit Industries: www.adafruit.com
Sparkfun electronics: www.sparkfun.com

Python:
Micro Python documentation for micro:bit
https://microbit-micropython.readthedocs.io/en/v2-docs/#

Other References:
https://www.w3schools.com/python/default.asp
https://www.python.org/ (advanced full reference for Python)
https://wiki.python.org/moin/BeginnersGuide (still pretty advanced)